

TABLE II

MATHEMATICAL OPTIMIZATION MODELS AND SOLUTION METHODS IN *REPRESENTATION & RENDERING*. HALF OF THE PAPERS ARE IN THIS PHASE OF THE DATA VISUALIZATION PIPELINE. DIFFERENT COLORED SQUARES DENOTE MINOR CATEGORIES. THE ROW OF THE TABLE IS SORTED BY THE NUMBER OF PAPERS

	Constrained or Unconstrained		Continuous, Discrete or Mixed			Linear or Nonlinear		Single-Objective or Multi-Objective	
	Constrained	Unconstrained	Continuous	Discrete	Mixed	Linear	Nonlinear	Single-Objective	Multi-Objective
Solver									
Intelligent Optimization									
First Derivative Method									
Simple Heuristic									
Two-Stage									
Approximation									
Dynamic Programming									
Graph Theory									
Active Set									
Hungarian									
Second Derivative Method									
Numerical Algebra									
Grid Search									
Analytic Method									
Simplex									

Grid & Rectangle Layout
 Node-Link Graph Layout
 Geographic Map Layout
 Hierarchical Structure Layout
 Perception & Color
 Rendering

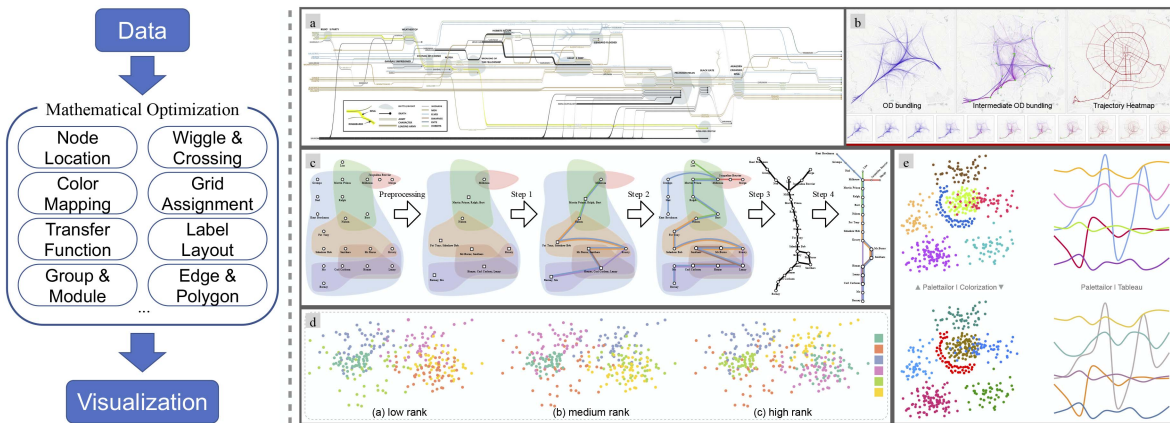


Fig. 5. The basic methodological paradigm and classical papers of *Representation & Rendering*. (a) Flow map layout [129] reduces confusion in story details. (b) Transitions between edge bundling and trajectory map improve faithfulness of actual movement paths [112]. (c) Metro map metaphor [142] visualizes set data as metro map by a multi-stage optimization layout. (d) Perception optimization [143] for various classes of color. (e) Palettator [144] integrates data aware manner into types of categorical data visualizations.

[80]. When applied to the layout of labels, it is usually necessary that individual labels do not overlap. Many works regard labels as rectangles and optimize layouts to determine the position of outlier labels in scatter plots [79], explore the relationship between labels [81], and facilitate tags search in documents [82].

Rectangle layout is ubiquitous in user interface layout because most computer programs use rectangle interfaces such as multi-display visual analysis [83], multiple user interfaces adaption [14], and web interface [84]. These works regard windows as rectangles and utilize constraints to limit their position in the process of layout optimization. Furthermore, ViSizer [85] resizes the visualization form for different devices automatically and solves the layout adjustment model by the previous method [86].

The grid layout is a type of polygon layout on a map. The goal is to arrange and align the grid on the map as tidily as possible. Many works have sought to simplify geographic maps as rectangular grids [87] and polygon grids [88], as shown in

Fig. 4(b). In these works, mapping areas as grids could be formulated as a typical set matching problem or linear assignment problem. The grid layout can also be used to present an overview of image pattern distribution [89], [90], clustered graphs [91], hypergraphs [92], euler diagrams [93], and combined with geographic information [94]. For example, a grid layout problem is commonly expressed as a linear assignment problem [95] to explore the image distribution pattern [89]. Moreover, Wu et al. [91] propose the multilevel technique of regional balancing in a cluster map layout by a multi-objective optimization model. Furthermore, the grid layout is also widespread used in the field layout such as conforming grid structures [96], surface grid mapping [97], and multiplanar grid reconstruction [98].

Graphical position relations are often used as objective functions or constraints. Since the rectangle position relation only considers horizontal and vertical directions, it results in linear constraints in the model. For polygon layouts, the problem is how to specify concise constraints which could describe